Vacuum Packaging Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Vacuum packaging machine

What it is used for

A vacuum packaging machine removes air from a package and seals it to help protect compatible products during storage and transport. Correct packaging settings and hygienic handling can improve shelf life, but vacuum packaging does not replace required food safety controls.

Main parts

- Vacuum chamber or external suction nozzle
- Vacuum pump and air extraction line
- Heat-sealing bar and sealing strip
- Sealing temperature and time controls
- Lid, gasket, and chamber locking arrangement
- Vacuum pressure indicator and emergency stop if fitted

Before you start

1. Read the operating instructions and confirm the machine is suitable for the product and packaging material.
2. Inspect the chamber, lid gasket, sealing bar, cables, and controls for damage or residue.
3. Check that the sealing strip is clean and the vacuum pump is ready for use.
4. Use clean, food-grade vacuum bags or approved packaging material of suitable thickness.
5. Ensure the product is at the required safe temperature and the package rim is clean and dry.
6. Keep liquids and powders within the machine's permitted packaging method and capacity.

How to use

1. Place the product inside the approved bag and leave sufficient space at the open end for sealing.
2. Position the bag opening flat across the sealing bar without folds or trapped product.
3. For chamber models, place the bag inside the chamber and close the lid securely.
4. Select the required vacuum level, sealing temperature, and sealing time for the packaging material.
5. Start the cycle and allow the machine to evacuate air and complete the seal automatically.
6. For external models, position the suction nozzle according to the machine instructions and keep the bag opening correctly aligned.
7. Wait until the cycle finishes and the chamber returns to safe opening conditions.
8. Remove the package and inspect the seal for continuity, wrinkles, contamination, or leaks.
9. Check the vacuum and seal quality using the approved inspection method.
10. Switch off the machine and clean product-contact surfaces and the sealing area after use.

⚠ Safety

- Keep fingers away from the hot sealing bar during and immediately after operation.
- Never bypass lid interlocks or force open a chamber under vacuum.
- Use only packaging materials approved for the machine's temperature and sealing method.
- Keep electrical components dry and disconnect power before cleaning or maintenance.
- Do not package hazardous chemicals or incompatible materials unless the machine and procedure are specifically approved.
- Follow food safety requirements for temperature control, sanitation, storage, and shelf life.

Care & storage

- Clean the chamber, lid, gasket, and sealing bar after use according to the manufacturer's instructions.
- Inspect the gasket, sealing strip, vacuum hose, and pump for wear or damage.
- Replace worn sealing elements with compatible approved parts.
- Maintain the vacuum pump and record cleaning, maintenance, and seal checks as required.

Common mistakes to avoid

- A wet or contaminated bag opening can cause an incomplete seal.
- Incorrect sealing temperature or time can weaken or burn the packaging material.
- Overfilling the bag can trap product in the seal and cause leaks.
- Assuming vacuum packaging makes food sterile or safe at room temperature can cause foodborne illness.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.